

History of Biological Classification

Carolus Linnaeus (1707-1788), Father of Taxonomy and Founder of Modern Systematics, introduced the system of hierarchical classification. In the 19th and 20th centuries numerical taxonomy and phylogenetic classification emerged.

Classification

Phylogenetic (Cladistic) Classification

It is an evolutionary classification based on how a common ancestry was shared. Cladistic classification summarises the 'genetic distance' between all species in the 'phylogenetic tree'. In cladistic classification characters such as **analogous characters** (characters shared by a pair of organisms due to convergent evolution e.g. wings in sparrows and patagia (wing like structures) in flying squirrels) and **homologous characters** (characters shared by a pair of organisms, inherited from a common ancestor e.g., wings of sparrows and finches) are followed/taken into consideration. **Ernst Haeckel** introduced the method of representing phylogeny by 'trees' or branching diagrams.

1.6 LEVELS AND HIERARCHY OF CLASSIFICATION

Human beings are not only interested in knowing more about different kinds of organisms and their diversities, but also the relationships among them. This branch of study is referred to as **SYSTEMATICS**. Systematics is the branch of science that deals with the vast diversity of life. It also reveals the trends and evolutionary relationships of different groups of the organisms. These relationships establish the phylogeny of organisms. A key part of systematics is taxonomy. Taxonomic **hierarchy** includes **seven obligate categories** namely **kingdom, phylum, class, order, family, genus** and **species**, and other intermediate categories such as subkingdom, grade, division, subdivision, subphylum, superclass, subclass, superorder, suborder, super family, subfamily, subspecies, etc. If we consider the three Domain classification, we have eight obligate categories.

Linnaeus was the first taxonomist to establish a definite hierarchy of taxonomic categories called taxa (singular: taxon) like kingdom, class, order, genus and species. Haeckel (1888) introduced the taxon **Phylum**. A species sometimes may have more subspecies, which show some morphological variations (intra-specific variations).

Taxonomic Categories: Nowadays the three **Domain** classification is followed. **CARL WOESE** and co-workers observed that many prokaryotes previously classified under 'Prokaryota/ Monera' are more closely related to the 'eukaryotes' and classified them under a separate Domain the **ARCHAEA**. This type of study is called '**MOLECULAR SYSTEMATICS**'.

Now there is a general agreement on the **THREE DOMAIN CLASSIFICATION** of the living organisms namely **DOMAIN-I** : BACTERIA, **DOMAIN-II** : ARCHAEA and **DOMAIN-III**: EUKARYA. (**NOTE**: *DOMAIN is a taxon higher than 'Kingdom'*).

Bacteria and **Archaea** represent two separate, distinct groups (called Domains). Thus, the modern taxonomists replaced the **Kingdom Monera** with two distinct Kingdoms - **BACTERIA** and **ARCHAEBACTERIA**. Studies on RNAs revealed that *Archaea* and *Eukarya* are more closely related to each other than to Bacteria. According to three domain system there are **six kingdoms** namely Eubacteria, Archaeobacteria, Protista, Plantae, Fungi and Animalia.

1. **Kingdom**: All multicellular, non-saprobic, heterotrophs are included in the kingdom **Animalia**/ Metazoa.
2. **Phylum**: It includes one or more classes. E.g. Phylum Chordata includes the classes Cyclostomata, Chondrichthyes, Osteichthyes, Amphibia, Reptilia, Aves and Mammalia, along with the protochordates. All these are based on common features such as presence of notochord, dorsal hollow nerve cord, pharyngeal slits, post-anal tail, etc., in some stage of the life history.
3. **Class**: It includes one or more related orders. E.g. The class Mammalia includes the orders Rodentia (**rats**), Chiroptera (**bats**), Cetacea (**whales**), Carnivora (**dogs**), Primates (**monkeys** and **apes-gorilla, gibbon** and **man**), etc.
4. **Order**: It includes an assemblage of one or more related families. E.g. The families Felidae and Canidae are included in the order **Carnivora** along with Hyaenidae (**hyenas**), Ursidae (**bears**), etc.
5. **Family**: It includes one or more related genera and can be distinguished from the other families by important characteristic differences. Family felidae includes the genus of cat (*Felis*), genus of leopard (*Panthera*), etc. The members of Felidae can be distinguished from those of Canidae (foxes, dogs, wolves). Adding the suffix-idae to the type generic name forms the name of the family e.g. adding the suffix 'idae' to the type Generic name '**Homo**' gives the name of the Family to which man belongs - **Hominidae**. The name of the subfamily can be coined by adding the suffix-inae to the type generic name (e.g. **Homininae**) and the name of the superfamily by adding the suffix-oidea (e.g. **Hominoidea**).
6. **Genus**: It is a group of related species, resembling one another in certain characters e.g. *Panthera leo* (lion), *Panthera tigris* (tiger) and *Panthera pardus* (leopard) belong to the genus *Panthera*.
7. **Species and Subspecies**: Species is the **basic unit** of classification in the hierarchical taxonomic system. Species is a group of similar organisms sharing a '**common gene pool**' and interbreeding freely, producing 'fertile' offspring. A species occurs in the form of many interbreeding groups called 'populations'.